

bam-processing

bam-processing extracts feature tables from a single indexed BAM input. The CLI accepts one .bai file and writes a DuckDB database with record, motif, methylation, and sample tables.

Install with uv

```
uv sync
```

On Apple Silicon, Metal support is optional and requires the macOS extra:

```
uv sync --extra macos
```

If you set `JAX_PLATFORMS`, use `METAL` for Apple GPU runs or `cpu` for CPU runs.

Run the CLI

```
uv run bam-processing /path/to/sample.bam.bai
```

Example with Metal:

```
JAX_PLATFORMS=METAL uv run --project bam_processing bam-processing /path/to/sample.bam.bai
```

Optional metadata and output settings:

```
uv run bam-processing /path/to/sample.bam.bai \
  --sample-name sample-01 \
  --age 64 \
  --group als \
  --output-path /tmp/sample-01.duckdb
```

Run with Docker

```
docker build -t bam-processing .
docker run --rm \
  -v /path/to/data:/data \
  bam-processing /data/sample.bam.bai
```